Structural and Spectroscopic Characterization of Novel Environmentally Sensitive Fluorescent Proteins

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ABSTRACT

This report will discuss the work done this summer on characterization of environment sensitive fluorescent proteins using spectroscopic and structure determination tools. We specifically characterized a salt concentration sensitive mutant of mRFPx, sRFPx, and a magnetic field sensitive protein, MagLOV. sRFPx exhibits additional blue shifted and red-shifted fluorescence spectra compared to the wild type mRFPx. We observed the far-red emission is dampened at high salt concentrations. We carried out detailed spectral characterization of sRFPx to understand the salt concentration and pH sensitivity of individual spectral bands of the chromophore. MagLOV is another fluorescent protein derived from an optogenetic photoswitchable protein AsLOV2. MagLOV exhibits a novel magnetic field effect (MFE) where under weak magnetic field, the fluorescence intensity of MagLOV decreases. As AsLOV2's fluorescence is closely tied to AsLOV2's structural photocycle, we hypothesize that MagLOV exhibits a structural change in response to weak magnetic fields (10-70mT). In this direction, we characterize the solution structure of MagLOV using SAXS. . By characterizing these environment sensitive fluorescent proteins, we hope to fuel future innovations in fields where fluorescent proteins are utilized, such as in vivo probes, or optogenetics.

INTRODUCTION—Environment sensitivity in Fluorescent proteins

Since their discovery in jellyfish in 1962, fluorescent proteins have been widely used as fluorescent probes. They can be tracked throughout a cell, and give good spatial information on whatever they're attached to or can detect. sRFPx is a mutant of the red fluorescent protein mRFPx. mRFPx has been widely used as a fluorescent probe in cells due to its ability to stay a monomer—making it easy to covalently attach to interesting structures—and its brightness (Bindels et al. 2016). It has since been engineered into a wide variety of other fluorescent proteins. sRFPx exhibits sensitivity to chloride anions. This has previously been observed in yellow fluorescent proteins, where fluorescence intensity will decrease in response to increasing NaCl concentration up to about 400 mM NaCl (Watcher and Reminton, 1999) (Galietta et al. 2001) These proteins hold significance as non-invasive probes to detect chloride concentrations in cellular environments, helping detect anomalous chloride levels which can lead to disease. sRFPx holds promise as another salt-sensitive fluorescent protein. We also hope to use sRFPx to investigate the structural differences between high and low salt conformations of these salt sensitive fluorescent proteins. Our work this summer in spectroscopic characterization supplements the structural information already generated by other members of the lab.

MagLOV is an engineered protein created through directed evolution of AsLOV2: a naturally occurring fluorescent blue light photoreceptor which has gained prominence in the field of optogenetics. MagLOV—a mutated AsLOV2 variant created by Hayward et al. at Calico

Biosciences—exhibits a novel property where its fluorescence will drop in response to exposure to a magnetic field. We call this Magnetic Field Effect MFE and have confirmed its existence in our own MagLOV protein expressed out of PET28a1 vectors.

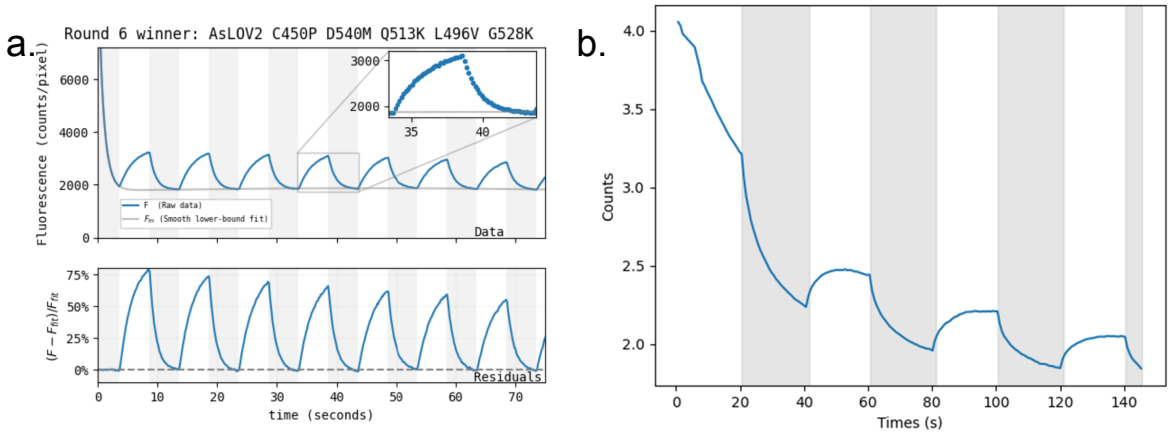


Fig 1. MFE measurements taken under 450nm light excitation. Y axis is the number of bright pixels detected by the experimental setup, which is indicative of how fluorescent the sample is. The way these pixels were counted was different between the two setups. As such absolute comparisons between these two experiments is difficult a) MFE measurements taken under 10 mT magnetic field (Hayward et al. 2024) expressed from PRSET vectors, b) MFE measurements carried out on our own protein expressed out of PET28a1+ vectors under 30-70mT magnetic field. Conducted on 8/13/24

This is of particular interest, as AsLOV2 has been previously used as an optogenetic tool, exhibiting structural change controllable by light (Lungu et al. 2012). This means scientists have been able to take advantage of this structural change, and tie certain biological functions to light. The canonical photocycle of AsLOV2 is understood to be an unfolding of a C terminus alpha helix termed the Jα Helix in response to the formation of a photoadduct between FMN and the Cysteine 450 residue on AsLOV2 (Lungu et al. 2012) (Zayner et al. 2012).

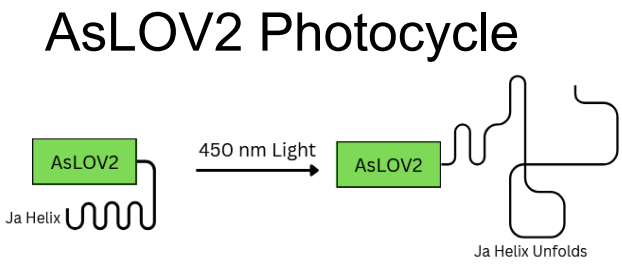


Fig 2. AsLOV2 Photocycle

As fluorescence activity is generally considered to be involved in the AsLOV2 photocycle (Swartz et al. 2001), it's possible MagLOV's magnetic modulation of its fluorescence

could trigger structural events. To get a better understanding of the structure of MagLOV, we designed and executed a small angle x-ray scattering (SAXS) experiment.

METHODS

Protein expression and purification of sRFPx mutant.

Protein samples were generated via expression in PET28a1 vector plasmids, transformed in BL21DE3 competent e.coli cells, and cultured through a primary culture where cells were grown, and a secondary culture where protein overexpression was encouraged through IPTG.

After lysing the cells with B-PER ® Bacterial Protein Extraction Reagent (Thermo Fischer), because our mRFPx construct is His Tagged, we performed an affinity based purification on Nickel-NTA agarose beads. Protein was stored in a final storage buffer of 30mM HEPES 150mM NaCl and flash frozen with dry ice and isopropyl alcohol and stored long term in -80 ℃ freezers.

Spectroscopic Characterization—

We used an infinite M1000 Tecan microplate reader, doing fluorescence emission scans on protein loaded into 96 well microplates. Protein was diluted to a final concentration of about 10uM in all wells, and in final data analysis variations in concentration were accounted for when necessary. Protein was diluted into citrate phosphate buffers at the desired pHs, and salt was added to these buffers as needed. Absorption data was gathered using a Nanodrop 2000c spectrophotometer.

Protein Purification of MagLOV

Protein samples generated last summer were his cleaved using Thrombin CleanCleave™ Kits from Millipore Sigma, and protein was concentrated to desired concentrations using Amicon® Ultra-15 Centrifugal Filter Units and Amicon® Ultra-4 Centrifugal Filter Units. Desalting and buffer exchange were performed in Thermo Fisher Zeba™ Spin Desalting Columns. Proteins were stored in a final storage buffer of 50mM HEPES 150mM NaCl 1mM Tcep before being flash frozen with dry ice and isopropyl alcohol and stored long term in -80 ℃ freezers.

SAXS studies with MagLOV

SAXS—or small angle X-ray scattering—is a technique which scatters an incident beam of X rays against the electron density around molecules present in a given sample solution. This scattering pattern can be used to give us data on the structure of our protein. It will provide us with a low resolution electron density map, along with information on the molecular size and degree of flexibility.

MagLOV samples for SAXS were stored at 21 mg/ml concentration, in 50 mM HEPES, 150 mM NaCl, 1 mM Tcep buffer, pH 7.4. Before being run through SAXS, the protein was

injected into a SEC column, where the SEC running buffer was 50 mM HEPES, 150 mM NaCl, 5 mM DTT.

We attempted this experiment with two injection volumes: a 10 uL and a 40 uL injection volume. BSA solution was used as an SEC standard.

To extract information regarding the size and disorder of each protein conformation, we analyzed our SAXS data using the ATSAS suite (EMBL Hamburg and SASDB). To generate our electron density maps we'll use a combination of the tools present in the ATSAS suite, as well as DENSS (Grant 2018).

RESULTS

Absorption and Emission Spectroscopy captures pH and ion sensitivity of mRFPx and sRFPx.

The absorption peaks present in sRFPx are distinct from mRFPx. In mRFPx pH decreases the absorption of the chromophore across the board, whereas with sRFPx it qualitatively changes the location of peaks. mRFPx exhibits an emission band with maxima at 570 nm with a shoulder near 540 nm (Figure 3a and Figure 3b).

sRFPx exhibits different absorption spectra depending on the pH. At low pH4.0—sRFPx has one large absorption peak (Figure 3c), while at neutral pH—around pH 7.0—sRFPx has distinct absorption peaks at 500 nm, as well as a broad peak from about 570 nm to 600 nm (Figure 3d). We do not not notice any significant changes between the absorption spectra taken in the presence of 150 mM and 2000 mM NaCl for both pH 4.0 and pH 7.0 (Figure 3c and 3d).

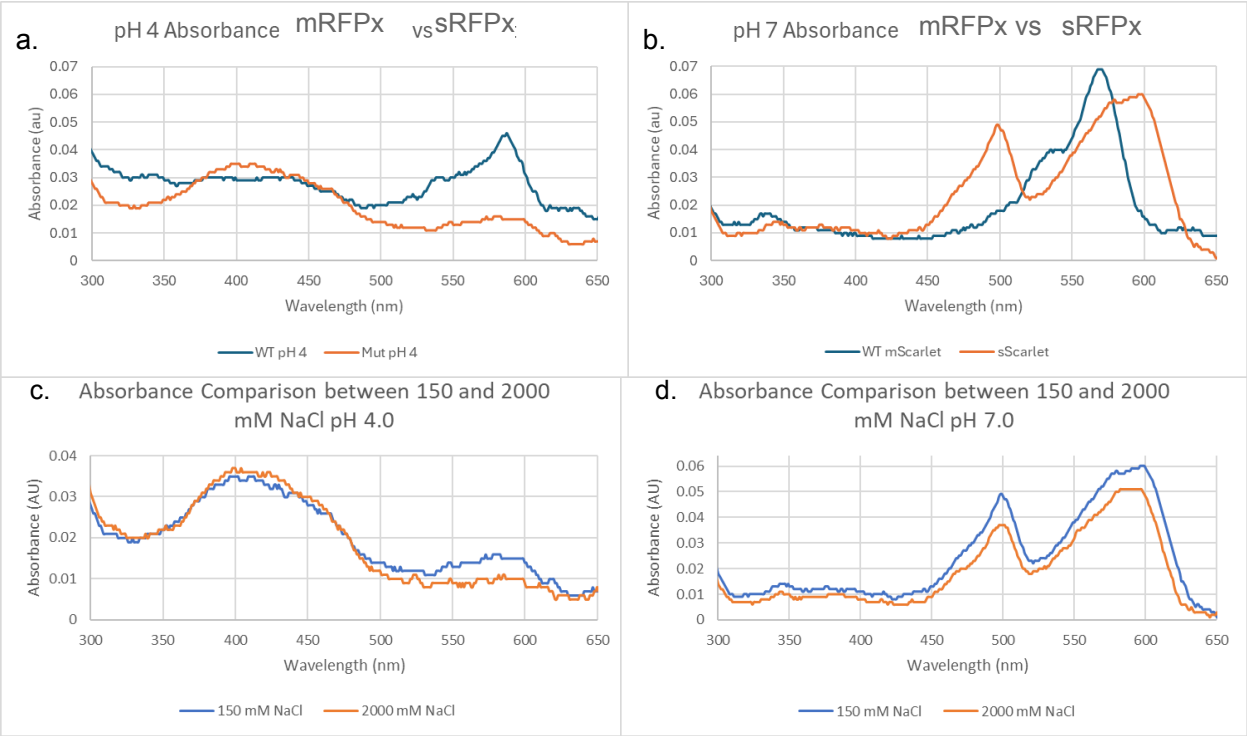


Fig 3. Absorption spectra of mRFPx and sRFPx at various pH and salt concentrations

From these absorption peaks, we took emission scans from the absorbance peaks we identified: 380, 500, 570 and 600 nms. At pH 4.0, 380 nm excitation emission was much greater than any of the other excitation wavelengths (Figure 4a). This reflects the fact that absorbance at 380 is much higher than anywhere else we took an emission scan (Figure 3c). At low pHs, there seem to exist two major emission peaks. One is present at 500 nm and the other at 615 nm. The latter peak—615—seems to be excitable by 380, 500, 570 and 600 nm light. At neutral pH, there seem to be three emission peaks, although two are very close to each other. The emission peak which likely peaks at around 510 nm is only present with 500 nm excitation. The other two are centered around 612 with excitation wavelengths of 500 nm and 570 nm (Figure 4b). There is another peak at around 620 nm which appears under excitations of 380 and 600 (Figure 4b).

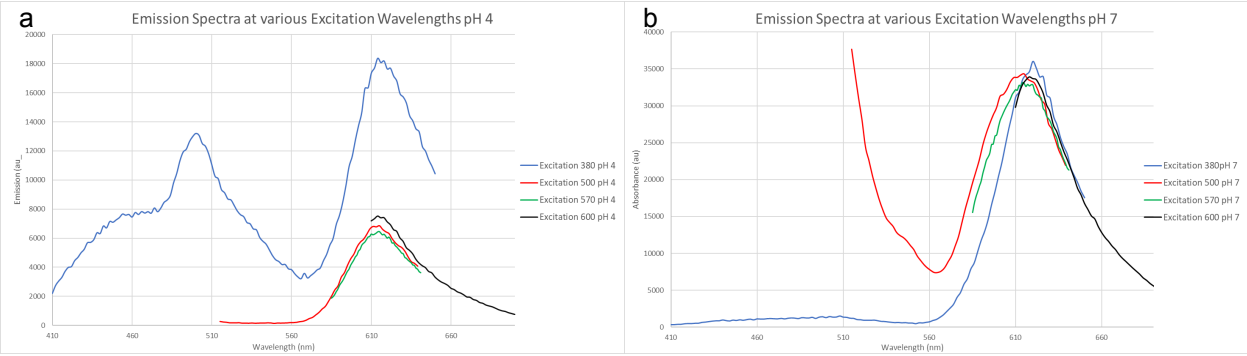


Fig 4. Emission scans for various excitation wavelengths at high and low pHs. All graphs taken at 250 mM NaCl.

This 620 peak shows a fair amount of salt sensitivity. With increasing salt sensitivity, the emission maximum recorded at an excitation wavelength of 600 nm shows that the emission maxima blue shifts by about 10 nm between 50 mM and 1 molar salt concentration (Figure 5)

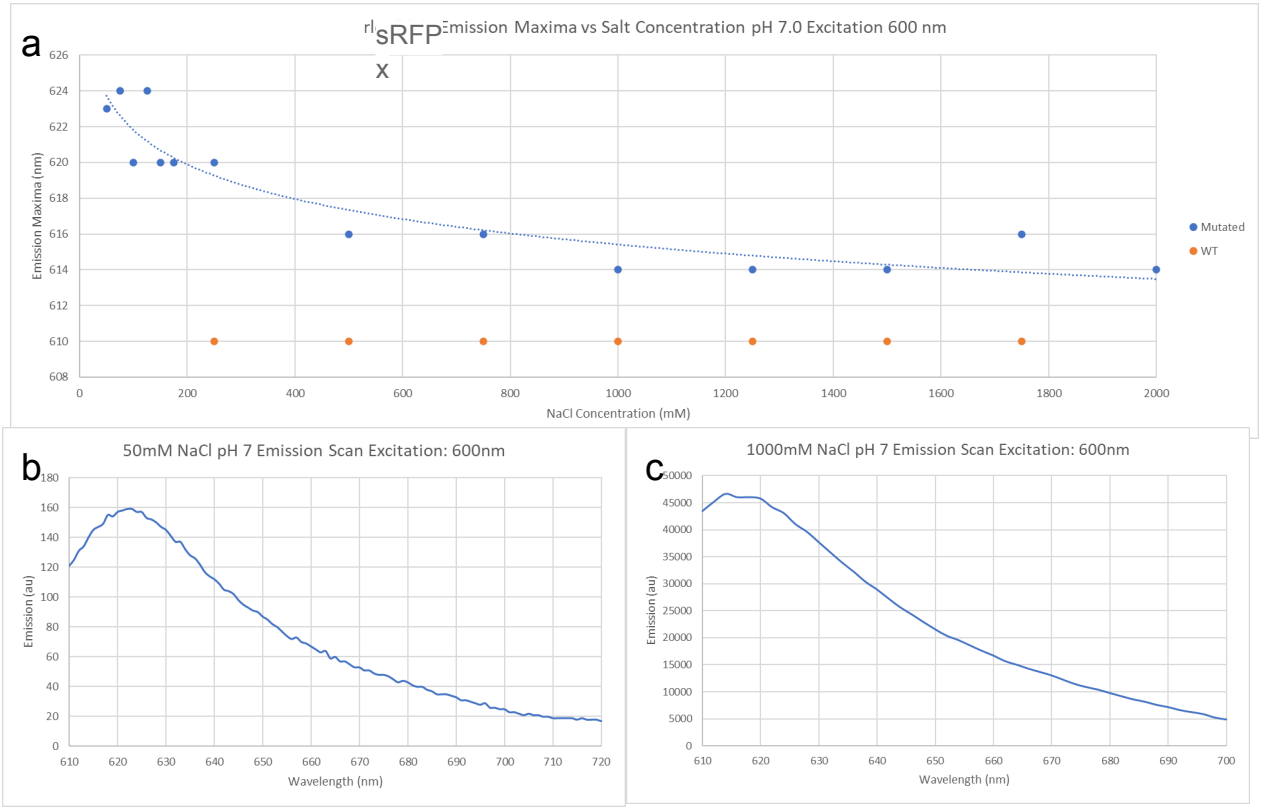


Fig 5a. Comparison of sRFPx emission maxima. The dotted line is a logarithmic trend line to guide the eye. The blue dots indicate sRFPx, while the orange dots indicate the wild type mRPGx Fig 5b and 5c. Emission spectra at 50 mM NaCl and 1 M NaCl.

Characterization of MagLOV solution structures by SEC-SAXS

SEC-SAXS is an experiment which first passes our protein sample through a size exclusion column to separate the solution out by size. These size excluded fractions are then fed into the SAXS cavity, where they scatter an incident beam of X rays against the electron density around molecules present in a given sample solution.

Between different injection volumes, there was no qualitative difference between the SEC traces—no new peaks appeared at 40ul compared to 10 ul. This is the SEC trace of MagLOV samples. The major peak at around image number 345 has a similar molecular weight as MagLOV to what we would expect from MagLOV based on its sequence (Figure 7).

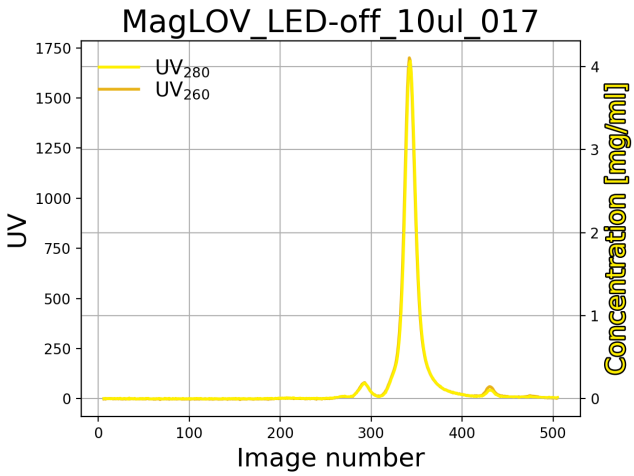


Figure 7. SEC trace of MagLOV solution. Image number is analogous to elution time in this case.

Next, we look at the Guinier analysis of the 345 image number peak. We see that data quality on the monomer peak is quite good, as it is so linear. As such, we can trust this SAXS data.

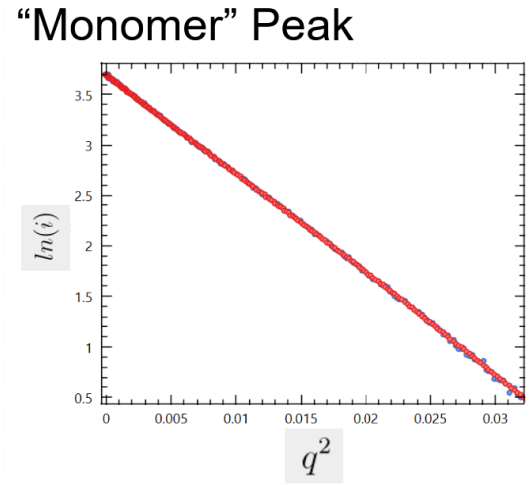


Figure 8. Guinier Analysis of MagLOV data averaged from image numbers 345-350 on 10 uL injection volume.

When fitted to an alphafold prediction of the rigid structure of MagLOV, the ab initio model of MagLOV generated from our SAXS data contains large regions of extra electron density (Figure 9).

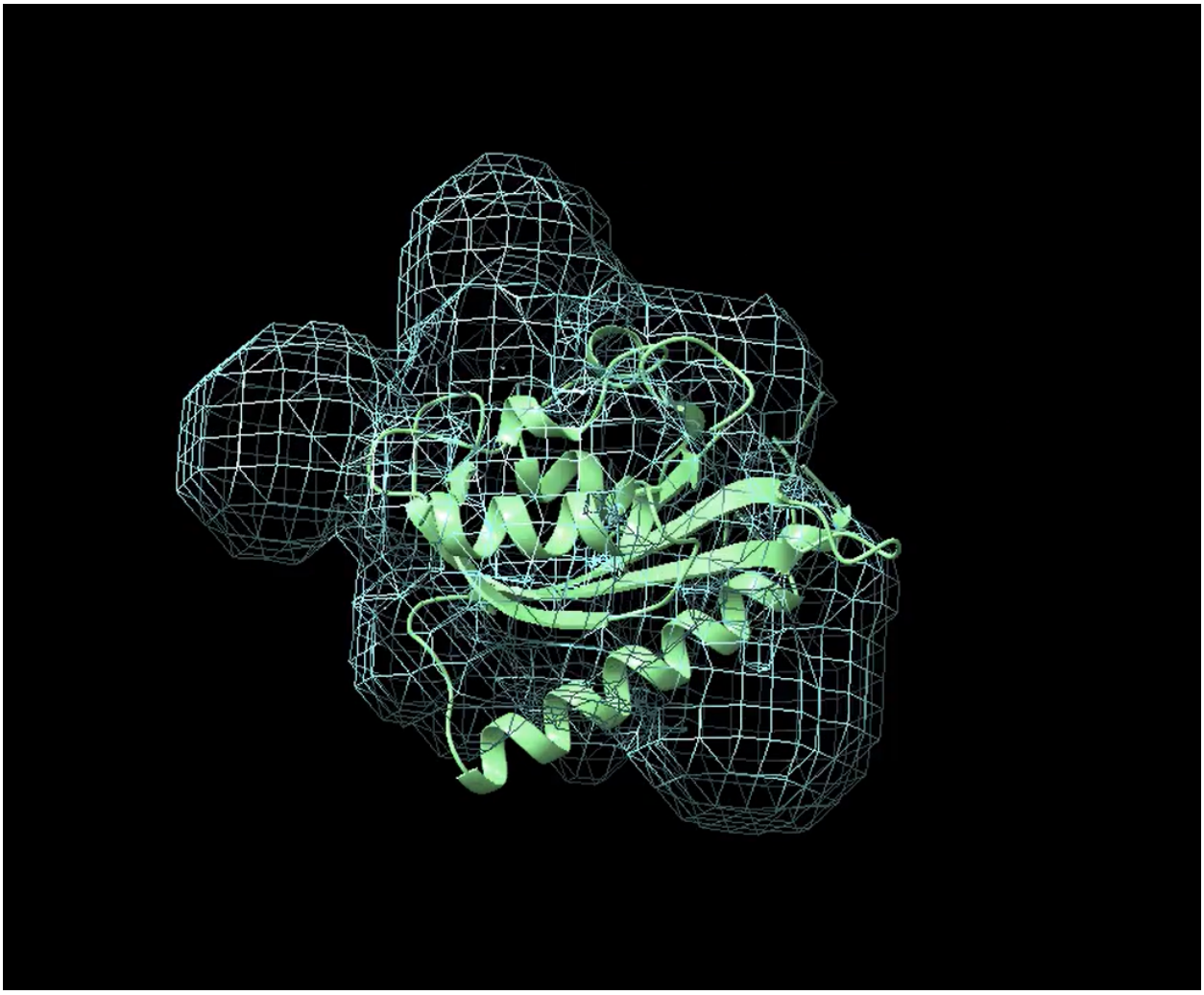


Figure 9. A prediction for MagLOV's structure is fitted to an ab initio electron density map model generated from SAXS data from the 345 image number peak

DISCUSSION

MagLOV shows increased degree of flexibility compared to AsLOV2

The extra electron density detected by SAXS is quite interesting (Figure 9). This could be an indication of data artifacting, although with how well the model fits into the center region of the electron density map, and the clean Guinier plot, this is hard to believe. We instead tentatively hypothesize that this indicates a high degree of flexibility with MagLOV's structure.

Further MD simulation, spectroscopic and structural studies could be performed to precisely solve the structure of MagLOV under the influence of various environmental stimuli such as magnetism or light.

Comparison between solution environment sensitivities between wild type and mutant mRFPx

The presence of three separate emission maxima in pH 7.0 and two separate emission maxima in pH 4.0 (Figure 4) indicates the existence of an equal number of chromophores within solution. Whether this means a single sRFPx molecule has three distinct chromophores, or if this means the sRFPx chromophore exists in three different electronic configurations remains to be seen.

The disappearance of the 624 nm emission peak which generates from 600 nm light excitation is our best evidence of salt sensitivity in sRFPx's chromophores (Figure 5a). This sensitivity is not found in mRFPx. This small blueshift proves sRFPxs capability as a salt sensing fluorescent protein.

CONCLUSION

Now that baseline characterization has been finished in both of these proteins, future work in the fields of structural, electronic and spectroscopic characterization can be performed. Questions about how sRFPx's electronic environment changes in response to salt can be raised, and perhaps answer questions on how salt concentration dependent FPs work in general. The determination of MagLOV's structure as it changes with light and magnetism could lead to light and magnetism controlled cellular function, as its precursor AsLOV2 was used; potentially opening a new field of magnetically controllable proteins.

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